

Institute of Science and Technology (IST)
B.Sc (Hons.) in CSE Part-1, 2nd Semester Model Test Examination
CSE-510223: Discrete Mathematics
Time Duration: 3 hours Full Marks: 80

[N.B. -- The figures in the right margin indicates full marks. All parts including subsections must be answered sequentially]

Answer any four of the following

1. a) What is disjunction and conjunction? Give truth table and example. 4
- b) Identify proposition for the following examples:- 3
 - i. Rio De Janeiro is the city of Mexico.
 - ii. What are you doing?
 - iii. Hello, Student.
 - iv. $Q < 9$
 - v. $1+1=2$
 - vi. $x+2=z$
- c) Differentiate i) implication and bi-implications. And ii) one to one and onto function. 4
- d) What is logical equivalence? Show that $(p \leftrightarrow q)$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent. 4
- e) Let $A_i = \{i, i+1, i+2, \dots\}$. Then $\bigcup_{i=1}^n = ?$ and $\bigcap_{i=1}^n = ?$ 5

2. a) What is a rule of inferences? 2
- b) What is the truth value of $\exists x P(x)$, where $\exists x P(x)$ is the statement " $n^2 > 10$ " and the universe of discourse consist of positive integers not exceeding 4? 5

- c) Define cardinality of a set, infinite set and power set. What is the power set of the empty set and of the set $\{2, 3, 4\}$? 6
- d) Let f be a function from $\{a, b, c, d\}$ to $\{10, 15, 20, 25\}$ with $f(a)=20, f(b)=15, f(c)=25$ and $f(d)=10$. Is f a bijection? 3
- e) Indirectly prove that if n is an integer and $3n+2$ is odd, then n is odd. 4

3. a) Define relation with example. Following are the relations on $\{1, 2, 3, 4\}$:
 - $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$
 - $R_2 = \{(1,1), (1,2), (2,1)\}$
 - $R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$
 - $R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$
 - $R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$
 - $R_6 = \{(3,4)\}$

Which of these relations are reflexive, symmetric, anti-symmetric and transitive?

- b) How many different bit string of length eight either start with a 1 or end with the two bits 00? 4

- c) State the principles of mathematical induction. Prove that $\sum_{j=0}^n ar^j = a + ar + ar^2 + \dots + ar^n = \frac{ar^{n+1} - a}{r-1}$ when $r \neq 1$ where n is a non-negative integer 2+4=6

Handwritten notes: $\{2, 3\}, \{3, 4\}, \{4\}, \{2, 3, 4\}, \{2, 4\}, \{2, 3\}$

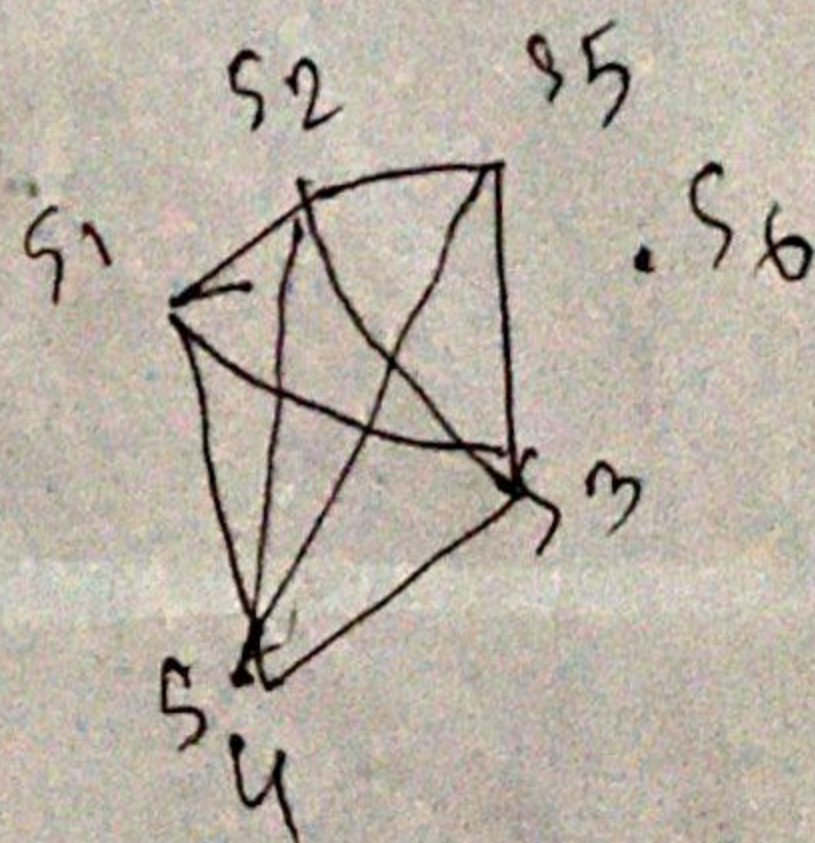
Handwritten notes: $\{3, 4\}, \{2, 4\}, \{2, 3, 4\}$

Handwritten notes: 128 - 14

Handwritten notes: 10, 5, 6, 2, 3, 5, 1, 4, 1, 2, 3, 4, 5, 6

4. a) What is graph? What are the types of graph, brief with example? 5
 b) Draw the precedence graph for the following expressions:- 2
 $S_1: a:=0$
 $S_2: a:=a+1$
 $S_3: b:=a+2$
 $S_4: c:=b+a$
 $S_5: d:=c+1$
 $S_6: e:=5$
 c) If G is a connected planar simple graph with e edges and v vertices, where $v \geq 3$, then prove that $e \leq 3v-6$. 5
 d) Identify Euler circuit, Euler path and Hamilton circuit, Hamilton path? 4
 e) Define chromatic number. What is the chromatic number of $K_{3,4}$ and K_5 ? 4
 5. a) Define m -ary and full m -ary tree with example. 4
 b) What is partial ordering? Construct a Hasse diagram for $(\{1, 2, 3, 4\}, \leq)$. 6
 c) Prove that there are at most m^h leaves in an m -ary tree of length h . 5
 d) What is spanning cost tree. Using Kruskal's algorithm find the minimum cost spanning route for the following graph. 1+4=5

	2	b	3	c	1	d
a						
3		1		2		5
e	4	f	3	g	3	h
4		2		4		3
	3		3		1	
i		i	k		m	



6. a) Construct half adder and full adder circuit using logical equations. 8
 b) Using Karnaugh maps to simplify the sum of products expressions 4
 (a). $xy\bar{z} + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}\bar{z}$.
 c) Construct circuit for the following expression using logic gates. 4
 i) $xz + \bar{x}y\bar{z}$ ii) $(x+z) + \bar{x}$
 d) Prove the absorption law $x(x+y) = x$ using the other identities of Boolean algebra. 2+1=3

